

Question Paper

Exam Date & Time: 01-Jan-2026 (09:30 AM - 12:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU

THIRD SEMESTER B.TECH. DEGREE MAKEUP

EXAMINATIONS - JANUARY 2026

PRINCIPLES OF DATA COMMUNICATION [IT_2124]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) A microwave transmitter has an output of 0.5 W at 5 GHz. Assume that this transmitter is used in a microwave communication system where the transmitting and receiving antennas are parabolas, each 1.5 m in diameter. (5)
- A)
- a. Calculate the gain of each antenna in decibels?
 - b. Considering antenna gain, what is the effective radiated power of the transmitted signal?
 - c. If the receiving antenna is located 30 km from the transmitting antenna over a free space path, find the available signal power out of the receiving antenna in dBm units.

Course Outcome: "CO1" / Cognitive Level: "Application"

- B) Apply the concept of Statistical Time Division Multiplexing (STDM) to design a packet-switched network that supports users with variable data transmission rates. Demonstrate how STDM can dynamically assign time slots based on user demand to achieve better bandwidth efficiency than synchronous TDM. (3)

Course Outcome: "CO3" / Cognitive Level: "Analysis"

- C) A slotted ALOHA network transmits 200-bit frames using a shared channel with a 200-kbps bandwidth. Find the throughput if the system (all stations together) produces (2)
- a. 1000 frames per second
 - b. 500 frames per second

Course Outcome: "CO4" / Cognitive Level: "Analysis"

- 2) A sender transmits the data 1101011011 with the generator polynomial 1011. At the receiver's end, the received data is 1101011001. Determine if there is an error in the transmission using CRC. (5)

- A) Course Outcome: "CO2" / Cognitive Level: "Application"

- B) A channel is transmitting data at a rate of 1.5 Mbps using a binary signal (2 signal levels). Calculate the minimum bandwidth required using the Nyquist theorem. (3)

Course Outcome: "CO1" / Cognitive Level: "Application"

- C) Tabulate the advantages and limitations of FDM in modern radio broadcasting systems. (2)

Course Outcome: "CO3" / Cognitive Level: "Application"

- 3) During transmission, a long sequence of bits contains the pattern 01111110, which is also used as (5)

A) the HDLC flag.

1. Explain how bit stuffing prevents the receiver from mistaking this data pattern as a frame boundary.

Give an example of how a data sequence is modified during bit stuffing and later restored at the receiver.

Course Outcome: "CO3" / Cognitive Level: "Analysis"

B) A Stop-and-Wait ARQ system transmits 1000-bit frames over a 1 Mbps link. (3)
The round-trip time (RTT) is 10 ms.
Calculate the efficiency of the system.

Course Outcome: "CO2" / Cognitive Level: "Application"

C) Calculate the baud rate and bandwidth for a signal transmitting at 10 Mbps, The value of $r = 0.35$ (for 16 QAM). (2)

Course Outcome: "CO5" / Cognitive Level: "Application"

4) In a network consisting of six LANs (LAN A, LAN B, LAN C, LAN D, LAN E, and LAN F), several source routing bridges (B1, B2, B3, B4, B5) connect these LANs. The initial spanning tree includes only bridges B1, B2, and B3, while B4 and B5 are backup bridges. (5)

- A)
1. Construct the Single Route Broadcast (SRB) frame path when a source station S1 on LAN A wants to discover the route to S2 on LAN F.
 2. Show the paths taken by the All-Route Broadcast (ARB) frames returned by S2.
 3. List all possible routes from S1 to S2 based on the All-Route Broadcast.
 4. Calculate the total number of LAN frames required to learn all possible routes between S1 and S2.

Course Outcome: "CO4" / Cognitive Level: "Application"

B) A transmission system is transmitting a file of size 10 MB over a channel with a bit rate of 1 Mbps. (3)
Calculate the time taken to transmit the file.

Course Outcome: "CO1" / Cognitive Level: "Application"

C) A TDM system multiplexes 20 voice channels, each sampled at 8 kHz using 8 bits per sample. (2)
1. Calculate the frame duration and bit rate of the TDM stream.
2. If a synchronization bit is added to each frame, find the new data rate.

Course Outcome: "CO5" / Cognitive Level: "Analysis"

5) An Ethernet LAN uses CSMA/CD. The maximum cable length gives a one-way propagation delay of 25 μ s between the two farthest stations. The physical layer bit rate is 10 Mbps. (5)

- A)
1. Calculate the round-trip propagation delay.
 2. Find the minimum frame size (in bits and bytes) required so that a transmitting station can detect a collision (assume collision must be detected before the frame transmission ends).
 3. If the network is upgraded to 100 Mbps but the same minimum frame size is used, explain whether collisions might go undetected.

Course Outcome: "CO4" / Cognitive Level: "Application"

B) A satellite link has $R=10$ Mbps, $RTT=500$ ms and frame size $L=10,000$ bits. (3)
1. Compute the efficiency of Stop-and-Wait ARQ.
2. Determine the minimum window size NNN required in Go-Back-N to reach at least 90%

efficiency.

Course Outcome: "CO3" / Cognitive Level: "Application"

- C) Calculate the propagation time and the transmission time for a 5-Mbyte message (an image) if the bandwidth of the network is 1 Mbps. Assume that the distance between the sender and the receiver is 12,000 km and that light travels at 2.4×10^8 m/s. (2)

Course Outcome: "CO1" / Cognitive Level: "Application"

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Event Name	Subject
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058-MIT BLR-MAKEUP EXAM-2025-2026

IT_2124:PRINCIPLES OF DATA COMMUNICATION

#	Course Outcome	Description	Marks
1	1	CO 1	13
2	2	CO 2	8
3	3	CO 3	13
4	4	CO 4	12
5	5	CO 5	4
Total			50

* **NA** - CO Not Selected

Event Name	Subject
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058-MIT BLR-MAKEUP EXAM-2025-2026

IT_2124:PRINCIPLES OF DATA COMMUNICATION

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	0	0(0%)
3	Application	11	38(76%)
4	Analysis	4	12(24%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
Total		15	50 (100%)